Hands-on Activity 11.1	
Basic Algorithm Analysis	
Course Code: CPE010	**Program:** Computer Engineering
Course Title: Data Structures and Algorithms	**Date Performed: Oct 6 2026**
Section: CPE21S1	**Date Submitted: Oct 6 2026**
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A. Output(s) and Observation(s)	

ILO A

ILO A: Measure the runtime of algorithms using theoretical analysis

Assess if two arrays have any common values.

```
bool diff(int *x, int *y){
    for(int i = 0; i < y.length; i++){
        if(search(x, y[i])!= - 1){
            return false;
        }
    }
    return true;
}
```

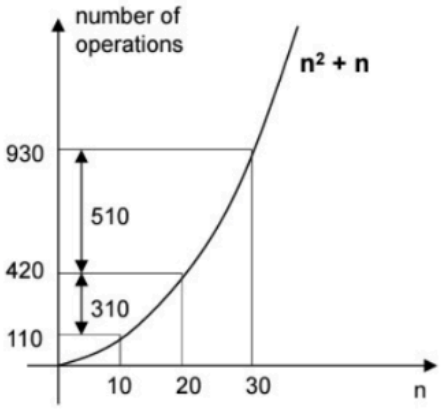
Performing worst case analysis:

- Let m = the length of array x.
- Let n = the length of array y.
- The loop in diff repeats n times.
- Each call to search requires m comparisons.

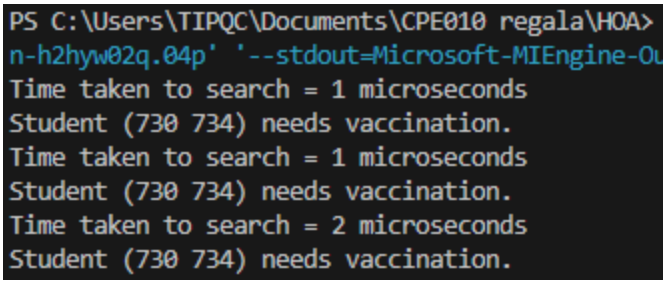
Given the above information, the total number of comparisons in the worst case is: _____

WORST CASE

Worst Case:

ILO A	• Assume that m = n. (The arrays are the same size) Provide an analysis of the graph for the worse case of the algorithm.  n (size of array) | number of operations 10 | 110 20 | 420 30 | 930 Analysis:
ANALYSIS	As we can see here the graph has the worst case is n(n+1) or n^2 + n as n grows the number of operations also quadratically increases.

ILO B			
INPUT SIZE	**EXECUTION SPEED**	**SCREENSHOT**	**OBSERVATION**
1,000	1 microsecond	Time taken to search = 1 microseconds Student (730 734) needs vaccination.	The search time is fast and only takes 1 microsecond
10,000	1 microsecond	Time taken to search = 1 microseconds Student (730 734) needs vaccination.	The search time is remains the same despite the larger input size
100,000	2 microsecond	Time taken to search = 2 microseconds Student (730 734) needs vaccination.	The search time increased to 2 microseconds but is still very fast

OUTPUT	
ANALYSIS	The code generates random students, sorts them, and uses binary search to find a specific student. It checks whether the student is vaccinated and returns whether vaccination is needed. The search time is measured using chrono for input sizes of 1,000, 10,000, and 100,000. The results show very fast execution because binary search has O(log n) time complexity.

B. Answers to Supplementary Activity

PROBLEM 1	**Problem 1**: Consider a program that returns true if the elements in an array are unique.
	
	`Algorithm Unique(A)`
	`  for i = 0 to n-1 do`
	`  for j = i+1 to n-1 do`
	`  if A[i] equals to A[j] then`
	`  return false`
	`  return true`

THEORETICAL	The space complexity is O(n^2 + n) bytes
EXPERIMENTAL	
ANALYSIS	

PROBLEM 2	**Problem 2**: Consider another program that raises a number x to an arbitrary nonnegative integer, n.
	
	`Algorithm rpower(int x, int n):`
	`1 if n == 0 return 1`
	`2 else return x*rpower(x, n-1)`
	
	`Algorithm brpower(int x, int n):`
	`1 if n == 0 return 1`
	`2 if n is odd then`
	`3  y = brpower(x, (n-1)/2)`
	`4  return x*y*y`
	`5 if n is even then`
	`6  y = brpower(x, n/2)`
	`7  return y*y`

THEORETICAL	
EXPERIMENTAL	

ANALYSIS	

C. Conclusion & Lessons Learned

Provide the following:
Summary of lessons learned
Analysis of the procedure
Analysis of the supplementary activity
Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?

IGNACIO	We learned about how Space complexity works and why it is important to make a code compact and efficient because in some microcontrollers some needs to have a more efficient and fast because some packages take too large to process, Theoretical Analysis should come first before experimental analysis although make one mistake then it wouldn't match up with our theoretical analysis and have to take a different approach if we want to get a certain outcome. For the supplementary activity, we compared the theoretical and experimental analysis and if we got the same for the theoretical and experimental analysis. It is a bit hard to understand for me since we are not measuring the time complexity anymore but rather than space it needs, It will take me a while to improve in this topic because this is very new to me.
REGALA	In this activity we learned about how algorithms can be analyzed using theoretical and experimental methods. We also learned that the number of inputs can affect the execution speed and that different levels have different efficiency even when performing similar tasks.

D. Assessment Rubric

T.I.P. S.O. 7 Rubric for Lifelong Learning (1)

Criteria	Ratings						Pts
Acquire and apply new knowledge from outside sources	**6 pts** **Excellent** Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently and applies knowledge learned into practice	**5 pts** **Good** Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently	**4 pts** **Satisfactory** Look beyond classroom requirements, showing interest in pursuing knowledge independently	**3 pts** **Unsatisfactory** Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	**2 pts** **Poor** Relies on classroom instruction only	**1 pts** **Very Poor** No initiative or interest in acquiring new knowledge	6 pts
Learn independently	**6 pts** **Excellent** Completes an assigned task independently and practices continuous improvement	**5 pts** **Good** Completes an assigned task without supervision or guidance	**4 pts** **Satisfactory** Requires minimal guidance to complete an assigned task	**3 pts** **Unsatisfactory** Requires detailed or step-by-step instructions to complete a task	**2 pts** **Poor** Shows little interest to complete a task independently	**1 pts** **Very Poor** No interest to complete a task independently	6 pts
Critical thinking in the broadest context of technological change	**6 pts** **Excellent** Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	**5 pts** **Good** Evaluate information from a variety of sources; formulates a clear and precise perspective	**4 pts** **Satisfactory** Analyze information from a variety of sources; formulates a clear and precise perspective.	**3 pts** **Unsatisfactory** Apply the gathered information to formulate the problem	**2 pts** **Poor** Gather and summarized the information from a variety of sources but failed to formulate the problem	**1 pts** **Very Poor** Gather information from a variety of sources	6 pts
Creativity and adaptability to new and emerging technologies	**6 pts** **Excellent** Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	**5 pts** **Good** Ideas are creative and adapt the new knowledge to solve a problem or address an issue	**4 pts** **Satisfactory** Ideas are creative in solving a problem or address an issue	**3 pts** **Unsatisfactory** Shows some creative ways to solve the problem	**2 pts** **Poor** Shows initiative and attempt to develop creative ideas to solve the problem	**1 pts** **Very Poor** Ideas are copied or restated from the sources consulted	6 pts

Total Points: 24

E. External References